A Wearable Brain-Computer Interface With Fewer EEG Channels for Online Motor Imagery Detection

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Background

Brain-computer interfaces (BCIs) allow users to control devices using brain signals commonly recorded through electroencephalography (EEG). EEG is a non-invasive technique that measures electrical activity through scalp electrodes.

Motor imagery (MI) BCIs detect brain activity produced during imagined movement. Since MI activates similar neural circuits as actual movement, they can enhance brain plasticity and aid in motor rehabilitation and recovery.

However, for these BCIs to be practical in real-world applications, wearability is crucial. Thus, recent research in this area has focused on improving portability and comfort while not sacrificing accuracy.

Problem Statement

- EEG-based BCIs tend to require multiple channels (electrodes) + other hindrances like wired connections and amplifiers → bulky systems not only decrease mobility and diminish their practical use, but also reduce user comfort (Figure 1)
- Decreasing the amount of electrodes used in the EEG cap → facilitates setup and portability, improving user convenience
  - May lead to performance degradation due to lower signal quality + unreliable decoding
- In order for the BCI to be used for rehabilitation, sufficient online performance is necessary → accuracy is required during real-time MI detection, resulting in immediate control and feedback

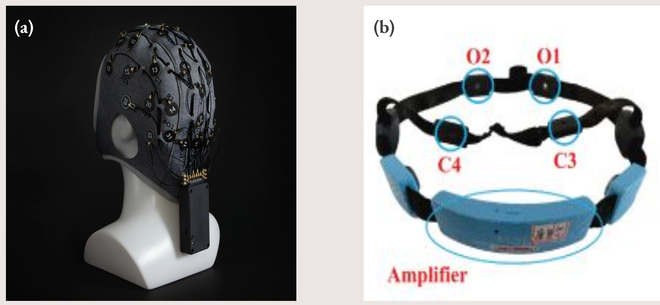

Figure 1. Comparison of EEG systems: (a) traditional, 32-channel cap and (b) lightweight, 4-channel head ring proposed in this study.

Therefore, the challenge with MI BCIs that rely on EEG is to find a way to reduce the channel number while maintaining precise and accurate online MI detection. **This study proposes and develops a wireless, wearable BCI system for online MI detection with a fewer number of channels.**

Methodology

- EEG recorded from 4 electrodes: O_1, O_2, C_3, C_4 (Figure 2), found in occipital lobe + motor cortex
- Commercial Neuroscan device used with same 4 electrodes to validate head ring performance
- 66 total participants in this study
  - Proposed wearable head ring system: 46 subjects
  - Comparative Neuroscan experiment: 20 subjects

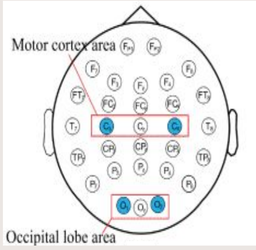

Figure 2. EEG electrode placements for study.

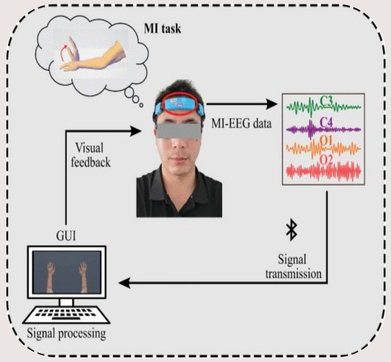

Figure 3. Closed-loop MI BCI system flowchart. EEG data from experiment was processed on a computer that had two hands on screen for real-time feedback.

- Subjects imagined wrist movement or stayed idle with 80 offline trials, then 60 online trials per session (Figures 3 & 4)
- 5-fold cross-validation (4 train/validation, 1 test) with 4 algorithms for signal processing (Table 1)
  - Analyzed using accuracy, F1 score, event-related desynchronization (ERD), and Shapley Additive explanations

Method	Features
EEGNet	extract + learn frequency and spatial filters
EEG-TCNet	adds temporal convolutional network (TCN)
TCNet-Fusion	strong feature map built with EEGNet + EEG-TCNet
ATCNet	adds multi-head self-attention to learn high-level features

Table 1. Algorithms used for processing and decoding the MI EEG brain signals.

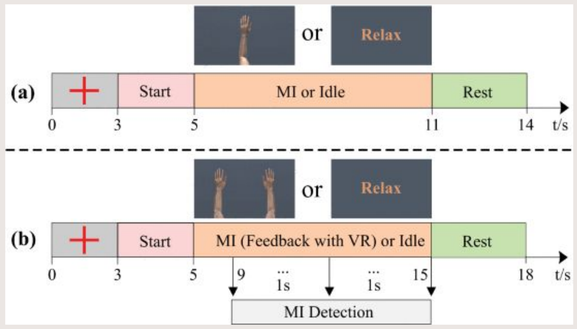

Figure 4. Experimental paradigms for (a) offline calibration and (b) online validation (MI detected once per second).

Results

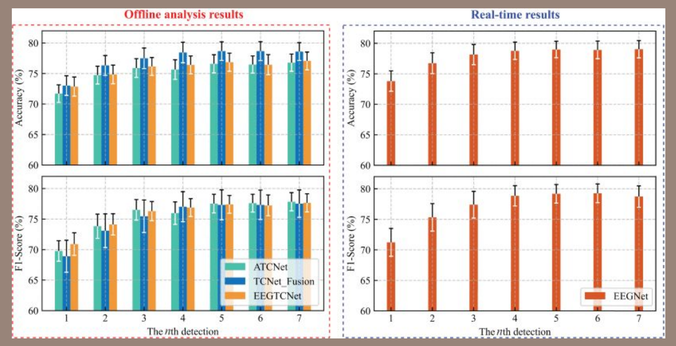

Figure 5. Offline (left) and online (right) classification accuracies and F1 scores. The EEGNet outperformed other methods, while also providing real-time results.

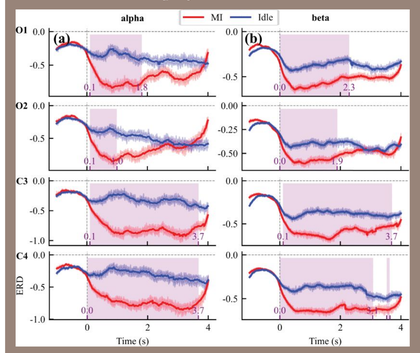

Figure 6. Average ERD amplitudes during MI and idle tasks for (a) alpha (8-13 Hz) and (b) beta (14-30 Hz) bands. ERD amplitudes were lower during MI for all 4 channels (known MI-EEG phenomenon).

Subject	Wearable headband		Wired Neuroscan	
	Offline	Online	Offline	Online
S1	92.22	76.19	96.25	89.05
S2	97.78	78.47	96.25	90
S3	82.22	73.68	87.62	73.33
S4	89.44	60.95	82.92	76.67
S5	89.44	90.95	91.25	68.1
S6	91.67	94.74	91.08	86.19
S7	90	71.77	94.92	89.05
S8	91.67	70.48	86.08	57.62
S9	95.56	89.52	87.5	86.19
S10	70	49.05	81.25	63.33
S11	85	78.85	82.5	87.14
S12	92.78	84.21	93.58	76.67
S13	92.22	93.33	97.5	89.52
S14	74.44	71.29	65.42	68.1
S15	83.33	86.12	93.58	81.9
S16	90.56	60.1	74.42	40.95
S17	92.22	87.5	93.75	74.76
S18	86.11	81.82	92.33	75.71
S19	93.89	80.48	92.42	70
S20	91.11	77.29	93.75	85.71
Mean ± STD	88.58 ± 6.65	77.84 ± 11.50	88.72 ± 7.91	76.50 ± 12.40

Table 2. Accuracy comparison of head ring and Neuroscan device.

Conclusion

- Accuracies and F1 scores were above 70% for all four methods during offline and online detection (Figure 5)
- Even with only 4 channels, this BCI system accurately represents movement-related brain activity (Figure 6)
- No significant difference in classification accuracies between head ring and Neuroscan device for both online and offline paradigms (Table 2)
- **With greater portability and no significant accuracy decline, wearable MI BCI system shows immense potential for motor rehabilitation and recovery**

References

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